

PROFESSIONAL SUMMARY

Computational biologist specializing in integrated multi-omic analysis and machine learning for cardiovascular biomarker discovery. Builds bioinformatics pipelines to mine large clinical cohorts to identify translational risk signatures. 5+ first-author publications (20+ total) and 2 pending U.S. patents from PhD work.

EDUCATION

- **PhD, Biomedical Engineering**, NYU Tandon School of Engineering 2023–2026, New York City, NY
- **Master of Science, Biomedical Informatics**, NYU Grossman School of Medicine 2022–2023, New York City, NY
- **Bachelor of Arts, Chemistry**, Reed College 2014–2018, Portland, OR

PROFESSIONAL EXPERIENCE

- **Graduate Research Assistant**, New York University Langone 2023–2026, New York, NY
 - Built and validated a whole-blood multiomic risk model for chronic coronary disease, integrating transcriptomic (n=646) and methylomic (n=732) profiling validated across two independent external cohorts (*Nature Communications*, 2026).
 - Characterized clonal hematopoiesis (CHIP) prevalence (23%) and its relationship to cardiovascular risk in 895 ISCHEMIA trial participants via whole-exome sequencing (*Circulation: Genomic and Precision Medicine*, 2025).
 - Developed and validated a platelet gene expression score (SLAP-GES, $r=0.76$, $P<0.001$) linking systemic lupus disease activity to vascular dysfunction (*JACC: Basic to Translational Science*, 2025).
 - Co-developed the Platelet Reactivity Expression Score (PRESS), an RNA-sequencing tool with an AUC of 0.81 for identifying hyperreactive platelet phenotypes; high-scoring patients had a 90% higher risk of cardiovascular events (*Nature Communications*, 2024).
- **Research Associate**, Oregon Health & Science University 2021–2022, Portland, OR
 - Led pre-clinical studies on catheter-based ultrasound therapy for peripheral artery disease and pulmonary embolism.
 - First-authored a book chapter on Ultrasound Imaging in Inflammation Research.
- **Research Assistant 2**, Oregon Health & Science University 2018–2021, Portland, OR
 - Characterized mechanisms and molecular drivers of aortic stenosis.
- **Advisory Board Member**, International Partners 2018–2022, Silver Spring, MD
 - Facilitated strategic planning and wrote grant proposals to fund community-driven programs.

AWARDS

- **2022**: ASE Foundation Early Career Investigator. *ASE Scientific Sessions*, Seattle, WA
- **2022**: Alan S. Pearlman Research Award. *ASE Scientific Sessions*, Seattle, WA

PATENTS

- **2025**: A Circulating Protein Signature of Platelet Reactivity Predicts Cardiovascular Risk. *U.S. Prov. App. No. 63/912,729*. With J.S. Berger, T.J. Barrett. Assignee: NYU.
- **2025**: A Platelet Transcriptomic Signature of Thromboinflammation Predicts Cardiovascular Risk. *U.S. Prov. App. No. 63/944,739*. With J.S. Berger, T.J. Barrett. Assignee: NYU.

SELECTED PUBLICATIONS

- [1] **Muller, Matthew**, M. G. Cornwell, S. Rajkumar, Z. Chen, *et al.*, “Whole blood epigenomic and transcriptomic characterization identifies vulnerable molecular subtypes of chronic coronary disease,” *Nature Communications*, 2026. Accepted.
- [2] **Muller, Matthew A.**, E. Luttrell-Williams, H. Bash, M. G. Cornwell, *et al.*, “Platelet Gene Expression in Systemic Lupus Erythematosus and Cardiovascular Health,” *JACC: Basic to Translational Science*, vol. 10, p. 101395, Dec. 2025.
- [3] **Muller, Matthew**, R. Liu, F. Shah, J. Hu, *et al.*, “Clonal Hematopoiesis of Indeterminate Potential in Chronic Coronary Disease: A Report From the ISCHEMIA Trials Biorepository,” *Circulation: Genomic and Precision Medicine*, p. e004921, 2025.
- [4] J. S. Berger, M. G. Cornwell, Y. Xia, **Muller, Matthew A.**, *et al.*, “A Platelet Reactivity Expression Score derived from patients with peripheral artery disease predicts cardiovascular risk,” *Nature Communications*, vol. 15, p. 6902, Aug. 2024.

SKILLS

- **Bioinformatics**: RNA-seq, scRNA-seq, WES/WGS, proteomics, methylomics, metabolomics, GWAS; DESeq2, Scanpy, GATK.
- **Data Science**: Machine learning (Scikit-learn, TensorFlow, PyTorch), Docker, Git.
- **Cloud Computing**: AWS, GCP, HPC/Slurm.
- **Languages**: Python, R, Bash, $\LaTeX$, SQL, Matlab.